ASSIGNMENT NO. – 6

AIM: Implement Bully and Ring algorithm for leader election.

Objective:

Coordinator that performs functions needed by other processes.

Outcome:

Current process P elects itself as a coordinator.

Explanation:

The **bully** algorithm is a type of **Election algorithm** which is mainly used for choosing a coordinate. In a distributed system, we need some election algorithms such as **bully** and **ring** to get a coordinator that performs functions needed by other processes.

Election algorithms select a single process from the processes that act as coordinator. A new process is selected when the selected coordinator process crashes due to some reasons. In order to determine the position where the new copy of coordinator should be restarted, the election algorithms are used.

It assumes that each process has a unique priority number in the system, so the highest priority process will be chosen first as a new coordinator. When the current use coordinator process crashes, it elects a new process having the highest priority number. We note that priority number and pass it to each active process in the distributed system.

The **Bully** election algorithm is as follows:

Let's assume that P is a process that sends a message to the coordinator.

It will assume that the coordinator process is failed when it doesn't receive any response from the coordinator within the time interval T.

An election message will be sent to all the active processes by process P along with the highest priority number.

If it will not receive any response within the time interval T, the current process P elects itself as a coordinator.

After selecting itself as a coordinator, it again sends a message that process P is elected as their new coordinator to all the processes having lower priority.

If process P will receive any response from another process Q within time T:

It again waits for time T to receive another response, i.e., it has been elected as coordinator from process Q.

If it doesn't receive any response within time T, it is assumed to have failed, and the algorithm is restarted

Practical 6A

```
BullyAlgo.j
ava:  import
java.io.*;
class
BullyAlgo
{
int
cood,ch,crash;
intprc[];
public void election(int n) throws IOException
{
BufferedReader br=new BufferedReader(new InputStreamReader(System.in));
System.out.println("\nThe Coordinator Has Crashed!");
int flag=1;
while(flag
==1)
{
crash=0;
for(int
i1=0;i1<n;i1++)
if(prc[i1]==0)
crash++;
if(crash==n)
{
System.out.println("\n*** All Processes Are Crashed ***"); break;
}
else
{
```

```
System.out.println("\nEnter The
Intiator"); int
init=Integer.parseInt(br.readLine());
if((init<1)||(init>n)||(prc[init-1]==0))
{
System.out.println("\nInvalid Initiator"); continue;
}
for(int i1=init-1;i1<n;i1++)
System.out.println("Process "+(i1+1)+" Called For Election");
System.out.println("");for(int i1=init-1;i1<n;i1++)
{
if(prc[i1]==0)
{
System.out.println("Process "+(i1+1)+ " Is Dead");
}
else
System.out.println("Process "+(i1+1)+" Is In");
}
for(int i1=n-
1;i1>=0;i1--)
if(prc[i1]==1)
{
cood=(i1+1);
System.out.println("\n*** New Coordinator Is "+(cood)+" ***");
flag=0;break;
}
}
}
}

public void Bully() throws IOException
{
BufferedReader br=new BufferedReader(new InputStreamReader(System.in));
System.out.println("Enter The Number Of Processes: ");
int
n=Integer.parseInt(br.readLine()
);prc=new int[n];
crash=0;
for(int
i=0;i<n;i++)
prc[i]=1;
coo
d=
n;
do
```

```
{
System.out.println("\n\t1. Crash A Process");
System.out.println("\t2. Recover A Process");
System.out.println("\t3. Display New
Cordinator");System.out.println("\t4. Exit");
ch=Integer.parseInt(br.readLine());
switch(ch)
{
case 1: System.out.println("\nEnter A Process To Crash"); int
cp=Integer.parseInt(br.readLine());
if((cp>n)||(cp<1)){
System.out.println("Invaid Process! Enter A Valid Process");
}
else if((prc[cp-1]==1)&&(cood!=cp))
{
prc[cp-1]=0;
System.out.println("\nProcess "+cp+ " Has Been Crashed");
}
else if((prc[cp-1]==1)&&(cood==cp))
{
prc[cp-
1]=0;
election
(n);
}
else
System.out.println("\nProcess "+cp+" Is Already Crashed");
break;case 2: System.out.println("\nCrashed Processes Are:
\n"); for(int i=0;i<n;i++)
{
if(prc[i]==0)
System.out.println(
i+1);crash++;
}
System.out.println("Enter The Process You Want To Recover"); int
rp=Integer.parseInt(br.readLine());
if((rp<1)||(rp>n))
System.out.println("\nInvalid Process. Enter A Valid
ID"); elseif((prc[rp-1]==0)&&(rp>cood))
{
prc[rp-1]=1;
System.out.println("\nProcess "+rp+" Has Recovered");
cood=rp;System.out.println("\nProcess "+rp+ " Is The New
Coordinator");
}
```

```
else if(crash==n)
{
prc[rp-
1]=1;
cood=r
p;
System.out.println("\nProcess "+rp+ " Is The New Coordinator"); crash--;
}
else if((prc[rp-1]==0)&&(rp<cood))
{
prc[rp-1]=1;
System.out.println("\nProcess "+rp+" Has Recovered");
}
else
System.out.println("\nProcess "+rp+" Is Not A Crashed Process");
break;case 3: System.out.println("\nCurrent Coordinator Is "+cood);
break; case 4: System.exit(0);
break;
default: System.out.println("\nInvalid Entry!"); break;
}
}
while(ch!=4);
}
public static void main(String args[]) throws IOException
{
BullyAlgo ob=new
BullyAlgo();ob.Bully();
}
}
```

Output:

```
itcgl09@itcgl09-OptiPlex-3070: ~/DS Practical/LP 6
itcgl09@itcgl09-OptiPlex-3070:~/DS Practical/LP 6$ javac BullyAlgo.java
itcgl09@itcgl09-OptiPlex-3070:~/DS Practical/LP 6$ java BullyAlgo
Enter The Number Of Processes:
4

        1. Crash A Process
        2. Recover A Process
        3. Display New Cordinator
        4. Exit
1

Enter A Process To Crash
4

The Coordinator Has Crashed!

Enter The Intiator
3
Process 3 Called For Election
Process 4 Called For Election

Process 3 Is In
Process 4 Is Dead

*** New Coordinator Is 3 ***

        1. Crash A Process
        2. Recover A Process
        3. Display New Cordinator
        4. Exit
```

```
itcgl09@itcgl09-OptiPlex-3070: ~/DS Practical/LP 6
itcgl09@itcgl09-OptiPlex-3070:~/DS Practical/LP 6$ javac BullyAlgo.java
itcgl09@itcgl09-OptiPlex-3070:~/DS Practical/LP 6$ java BullyAlgo
Enter The Number Of Processes:
4

        1. Crash A Process
        2. Recover A Process
        3. Display New Cordinator
        4. Exit
1

Enter A Process To Crash
4

The Coordinator Has Crashed!

Enter The Intiator
3
Process 3 Called For Election
Process 4 Called For Election

Process 3 Is In
Process 4 Is Dead

*** New Coordinator Is 3 ***

        1. Crash A Process
        2. Recover A Process
        3. Display New Cordinator
        4. Exit
```

Result:

You have executed all the processes

Date:	
Marks obtained:	
Sign of course coordinator:	
Name of course Coordinator :	